

Erythritol is a sweet antioxidant.

OBJECTIVE: Hyperglycemia, oxidative stress, and the onset and progression of diabetic complications are strongly linked. Reduction of oxidative stress could be of utmost importance in the long-term treatment of diabetic patients. The chronic nature of the disease calls for a mode of antioxidant intake that can be sustained easily, e.g., by the diet. Erythritol, a simple polyol, could be such a compound. It is orally available, well tolerated, and its chemical structure resembles that of mannitol, a well-known hydroxyl radical (HO*) scavenger. **METHODS:** We studied the antioxidant properties of erythritol in vitro and subsequently determined its antioxidant activity and its vasoprotective effect in the streptozotocin diabetic rat. **RESULTS:** Erythritol was shown to be an excellent HO* radical scavenger and an inhibitor of 2,2'-azobis-2-amidinopropane dihydrochloride-induced hemolysis but inert toward superoxide radicals. High-performance liquid chromatographic and electron spin resonance spectroscopy studies showed that the reaction of erythritol with hydroxyl radicals resulted in the formation of erythrose and erythrulose by abstraction of a carbon-bound hydrogen atom. In the streptozotocin diabetic rat, erythritol displayed an endothelium-protective effect and, in accordance with the in vitro experiments, erythrose was found in the urine of erythritol-consuming rats. **CONCLUSION:** Erythritol acts as an antioxidant in vivo and may help protect against hyperglycemia-induced vascular damage. Copyright 2010 Elsevier Inc. All rights reserved.